

Practical – Week9

1. Implement a file copy utility using low-level Linux file I/O system calls (open, read, write, lseek, close). Compare its performance with standard library functions (fopen, fread, fwrite).

```
GNU nano 8.7.1 lowlevel.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>
#include <time.h>
int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }
    int src = open(argv[1], O_RDONLY);
    if (src == -1) {
        perror("Source file error");
        return 1;
    }
    int dest = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (dest == -1) {
        perror("Destination file error");
        close(src);
        return 1;
    }
    // Find file size using lseek()
    off_t size = lseek(src, 0, SEEK_END);
    lseek(src, 0, SEEK_SET);
    char buffer[4096];
    ssize_t bytes;
    clock_t start = clock();
    while ((bytes = read(src, buffer, sizeof(buffer))) > 0) {
        ssize_t written = 0;
        while (written < bytes) {
            ssize_t n = write(dest, buffer + written,
                              bytes - written);
            if (n <= 0) {
                perror("Write error");
                close(src);
                close(dest);
                return 1;
            }
            written += n;
        }
    }
    clock_t end = clock();
    printf("Low-level copy time: %.6f seconds\n", (end - start) / CLOCKS_PER_SEC);
    return 0;
}
```

```
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ nano lowlevel.c
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ gcc lowlevel.c -o lowlevel
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ ./lowlevel input.txt output1.txt
File size: 10485760 bytes
Low-level copy time: 0.005918 seconds
```

```
#include <stdio.h>
#include <time.h>

int main(int argc, char *argv[]) {
    if (argc != 3) {
        printf("Usage: %s source destination\n", argv[0]);
        return 1;
    }
    FILE *src = fopen(argv[1], "rb");
    if (src == NULL) {
        perror("Source file error");
        return 1;
    }
    FILE *dest = fopen(argv[2], "wb");
    if (dest == NULL) {
        perror("Destination file error");
        fclose(src);
        return 1;
    }
    // Find file size using fseek()
    fseek(src, 0, SEEK_END);
    long size = ftell(src);
    fseek(src, 0, SEEK_SET);
    char buffer[4096];
    size_t bytes;
    clock_t start = clock();
    while ((bytes = fread(buffer, 1, sizeof(buffer), src)) > 0) {
        if (fwrite(buffer, 1, bytes, dest) != bytes) {
            perror("Write error");
            fclose(src);
            fclose(dest);
            return 1;
        }
    }

    if (ferror(src)) {
        perror("Read error");
    }
}
```

```
pawan_kumar@LAPTOP-SSDOPUQ2: ~/FileCopy
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ nano standard.c
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ gcc standard.c -o standard
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ dd if=/dev/zero of=input.txt bs=1M count=10
10+0 records in
10+0 records out
10485760 bytes (10 MB, 10 MiB) copied, 0.00920426 s, 1.2 GB/s
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ ls -lh input.txt
-rw-r--r-- 1 pawan_kumar pawan_kumar 10M Sep 27 07:33 input.txt
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ ./lowlevel input.txt output1.txt
File size: 10485760 bytes
Low-level copy time: 0.012839 seconds
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ ./standard input.txt output2.txt
File size: 10485760 bytes
Standard I/O copy time: 0.011981 seconds
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ time ./lowlevel input.txt output1.txt
File size: 10485760 bytes
Low-level copy time: 0.006772 seconds

real    0m0.010s
user    0m0.000s
sys     0m0.010s
pawan_kumar@LAPTOP-SSDOPUQ2:~/FileCopy$ time ./standard input.txt output2.txt
File size: 10485760 bytes
Standard I/O copy time: 0.007804 seconds

real    0m0.013s
user    0m0.000s
sys     0m0.013s
```

Report: The file was successfully copied using `open()`, `read()`, and `write()` system calls and `fopen()`, `fread()`, and `fwrite()` library functions. The `cmp` command verified that both output files matched the original, and the `time` command was used to compare their execution times.

2. Develop a program that redirects standard input and output using file descriptors and `dup2()`. Demonstrate how shells implement I/O redirection.

```

GNU nano 8.7.1                                redirect.c *
#include <stdio.h>
#include <fcntl.h>
#include <unistd.h>
#include <stdlib.h>

int main() {
    int fd_in, fd_out;
    // Open input file
    fd_in = open("input.txt", O_RDONLY);
    if (fd_in < 0) {
        perror("Input file error");
        return 1;
    }
    // Open output file
    fd_out = open("output.txt",
                  O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (fd_out < 0) {
        perror("Output file error");
        close(fd_in);
        return 1;
    }
    // Redirect standard input
    dup2(fd_in, STDIN_FILENO);
    // Redirect standard output
    dup2(fd_out, STDOUT_FILENO);
    close(fd_in);
    close(fd_out);
    // Execute a command using redirected I/O
    execlp("wc", "wc", "-l", (char *)NULL);
    perror("execlp failed");
    return 1;
}

```

```

pawan_kumar@LAPTOP-SSD0PUQ2:~/OS_Lab$ nano redirect.c
pawan_kumar@LAPTOP-SSD0PUQ2:~/OS_Lab$ nano input.txt
pawan_kumar@LAPTOP-SSD0PUQ2:~/OS_Lab$ cat input.txt
Hello
Welcome to Linux
Operating Systems

```